7.1.1 Where do sparse linear systems come from?

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Week 7-8 due Mar 2, 2025 01:52 IST

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going to say, "Let's make the problem a little bit simpler."

Let's say that the membrane is square. Okay? And secondly, let's assume that on

the boundary, the boundary condition is that the function takes on the value

zero. There's no displacement there. And then, what else can we do? Well, we know

from calculus that derivatives can be approximated using finite differences.

And what does that mean? Well, what it means is that this

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Reading assignment

0.0/1.0 point (graded)

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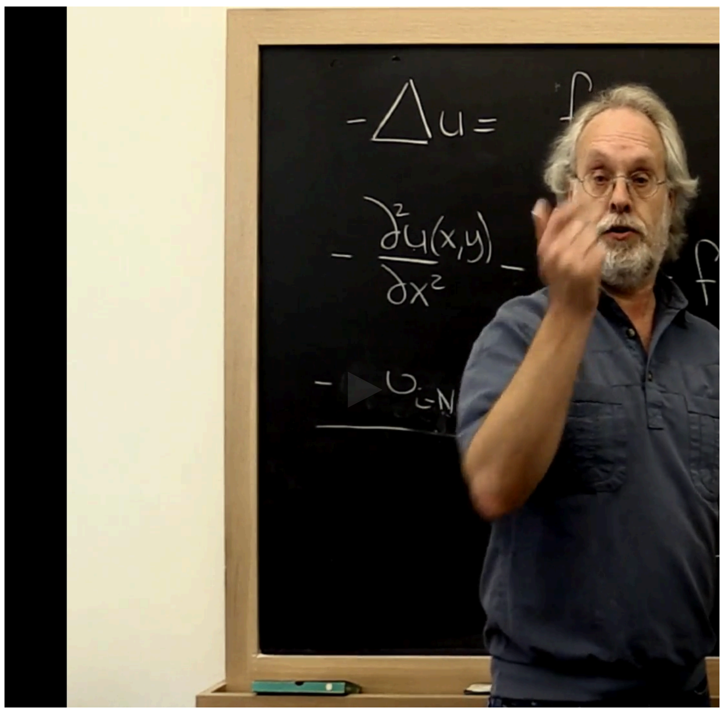

know that those points are equal to zero. Alright? So once we have renamed things here, we can look at this equation right here and say, "Oh, this is really upsilon_i-N, upsilon_i-1. There's a minus sign here. Then we get a plus 4 times upsilon_i, minus upsilon_i+1 and finally minus upsilon_i+N. And then this right here becomes a vector as well. Let's call that vector f. And let's say that the value at the i,j point is

3:42 / 3:42

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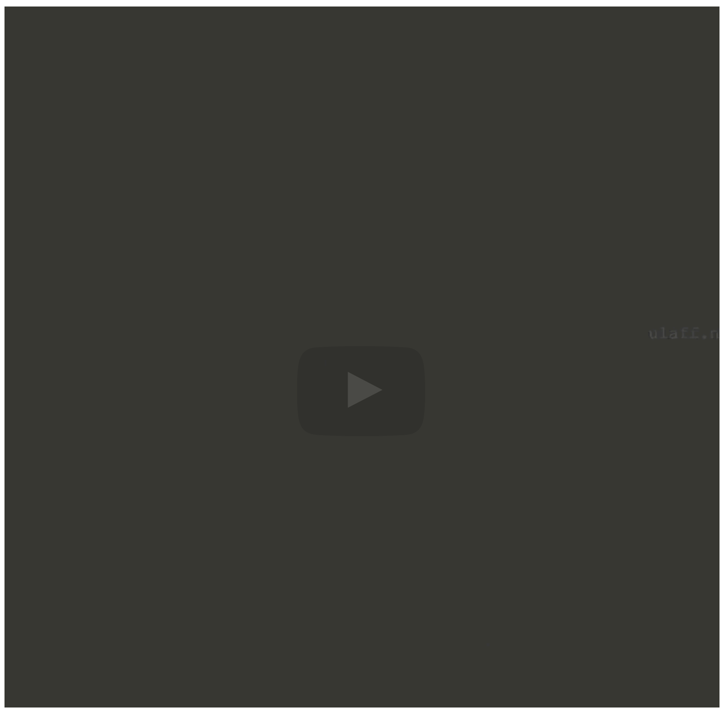

see, the point to the left is now equal to zero, so this one disappears, but we get a four and then another minus one, and a minus one. And notice that it's starting to look a little bit difficult here. If we put a bar here and a bar here, then it tells us where we got to the end of a line here. And what that really means is that all of the unknowns up to this point correspond to this first line. And then we start over again with another one. Okay? And what you

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Homework 7.1.1.1

1/1 point (graded)

The observations in this unit suggest the following way of solving

$$-\frac{\partial^2 u}{\partial x^2} - \frac{\partial^2 u}{\partial y^2} = f(x, y)$$

- Discretize the domain $0 \leq \chi, \psi \leq 1$ by creating an $(N+2) \times (N+2)$ mesh of points.
- An $(N+2) \times (N+2)$ array `U` holds the values $u(\chi_i, \psi_i)$ *plus the boundary around it.*
- Create an $(N+2) \times (N+2)$ array F that holds the values $f(\chi_i, \psi_j)$ (plus, for convenience, extra values that correspond to the boundary).
- Set all values in U to zero. This initializes the last rows and columns to zero, which captures the boundary condition, and initializes the rest of the values at the mesh points to zero.
- Repeatedly update all interior points with the formula

  $$U(i,j) =$$
  $$(h^2 \underbrace{F(i,j)}_{f(\chi_i, \psi_j)} + \underbrace{U(i,j-1)}_{u(\chi_i, \psi_{j-1})} + \underbrace{U(i-1,j)}_{u(\chi_{i-1}, \psi_j)} + \underbrace{U(i+1,j)}_{u(\chi_{i+1}, \psi_j)} +$$

  until the values converge.
- Bingo! You have written your first iterative solver for a sparse linear system.
- Test your solver with the problem where $f(\chi, \psi) = (\alpha^2 + \beta^2)\pi^2 \sin(\alpha\pi\chi)\sin(\beta\pi\psi)$.
- Hint: if `x` and `y` are arrays with the vectors x and y (with entries χ_i and ψ_j), then `mesh( x, y, U )` plots the values in `U`.

☑ done

Solution:

For complete solution see the notes .

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Answers are displayed within the problem

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You may yearn for an actual definition of a sparse matrix. Wilkinson, who

received the Turing Award for his work of numerical linear algebra, defines

a sparse matrix as a

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